CVPR JUNE 3-7, 2026 DENVER COLORADO

Project page

MUFASA: A Multi-Layer Framework for Slot Attention

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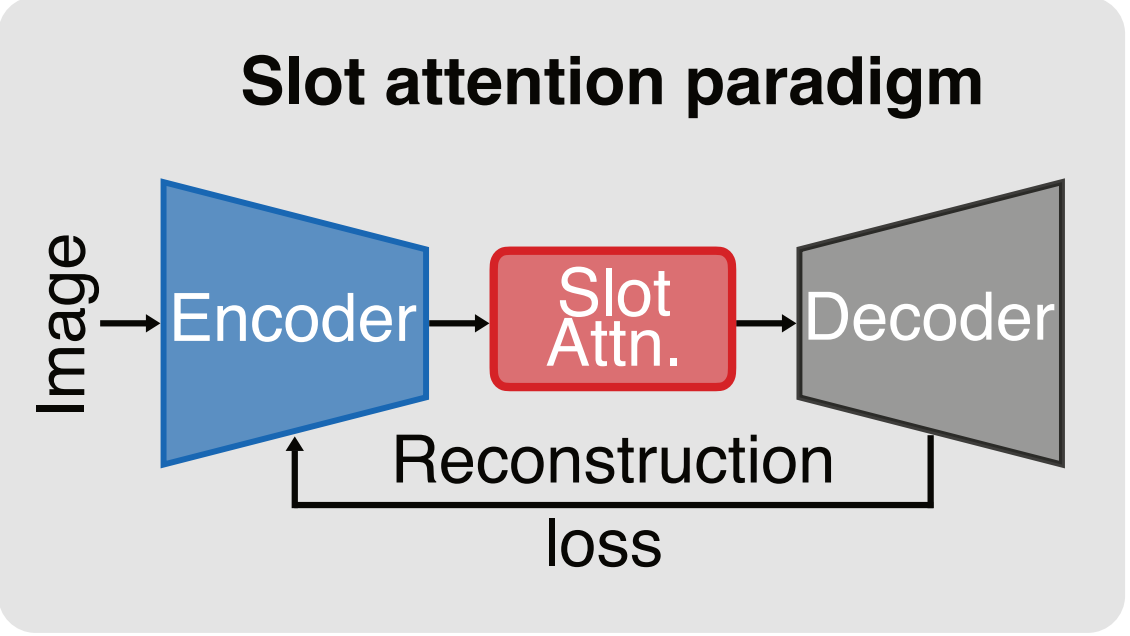
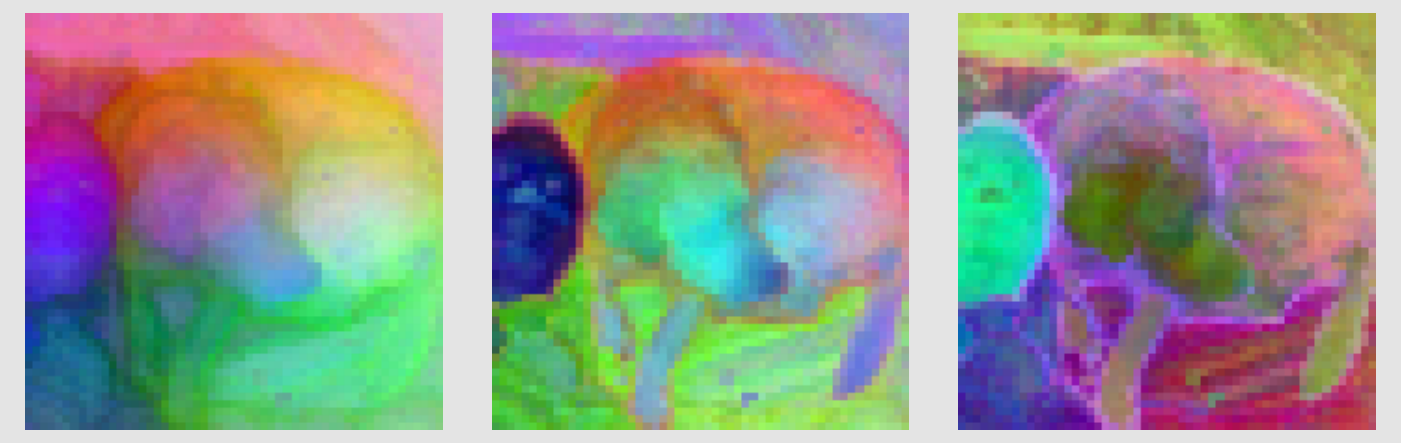
Motivation

- **Object-centric learning (OCL)** decomposes visual scenes into object-specific representations.
- **Slot attention (SA)** groups input features into a set of latent vectors (slots), which compete to bind to individual objects. Slot attention masks yield object segmentations.


Slot attention paradigm
Image
Encoder
Slot Attn.
Decoder
Reconstruction loss

- **Current approaches** like DINOSAUR [1] and SPOT [4] utilize **solely the last layer** of a pre-trained DINO ViT as input and reconstruction target.

DINO encoders learn a **representation hierarchy** across layers: shallow layers contain spatial information, deep layers encode increasingly rich semantics [3].


DINO features at layers 10, 11, and 12

Slot attention masks per layer

Layer-wise analysis

Especially deeper layers contain useful representations for SA:

- PCA reveals semantically meaningful and complementary information.
- Using these layers as input in SA yields object-aligned segmentations with varying level of details.

How can we effectively leverage this feature hierarchy to improve slot attention?

Contributions

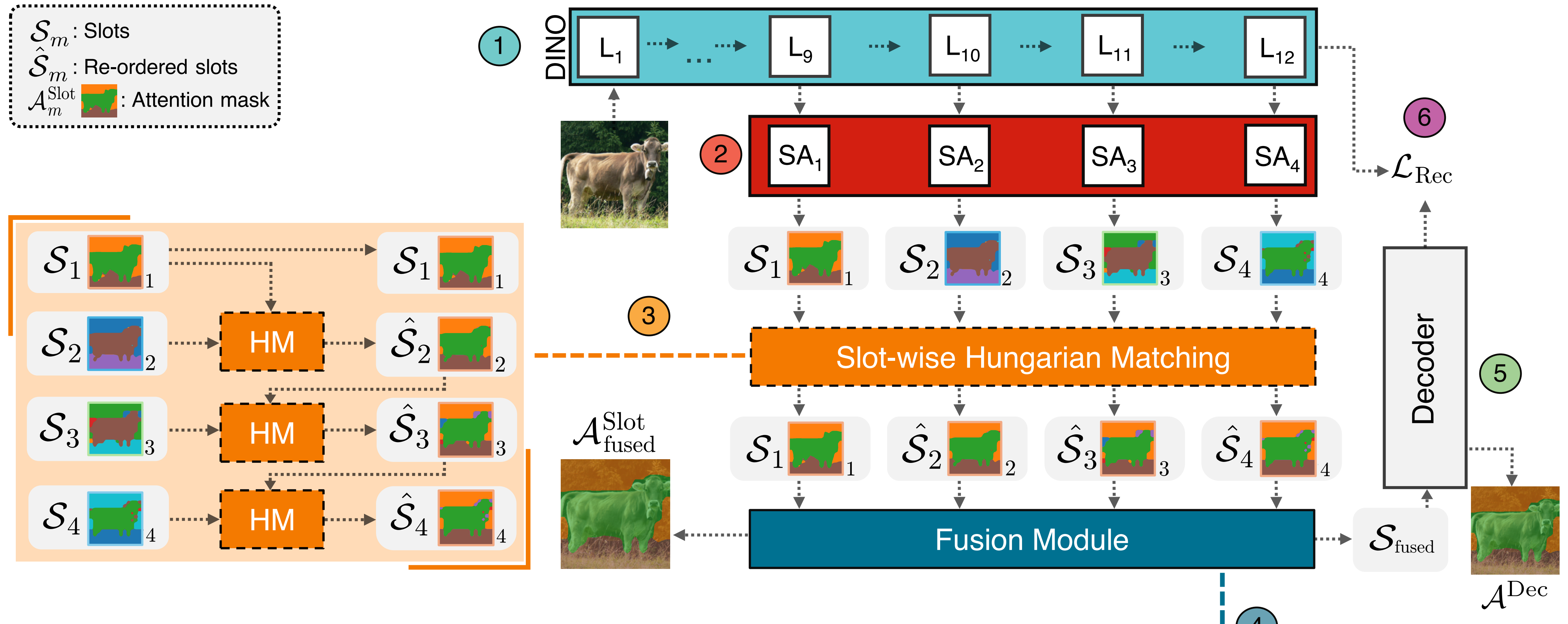
MUFASA: A novel framework for SA-based models that leverages the complementary features of multiple DINO layers for unsupervised object segmentation (UOS).

1. **M-Fusion:** Our fusion technique to effectively combine slots obtained at different layers into a unified representation.
2. **Plug-and-play:** MUFASA can easily be integrated into SA models utilizing DINO encoders, without introducing any additional supervision.
3. **Improved segmentation performance** across multiple datasets and models, setting a new SOTA in UOS when integrated into SPOT.
4. **Lightweight:** Faster training efficiency with minimal parameter and inference overhead.

References

[1] Seitzer et al., Bridging the gap to real-world object-centric learning, ICLR, 2023.
[2] Locatello et al., Object-centric learning with slot attention, NeurIPS, 2020.
[3] Amir et al., On the effectiveness of ViT features as local semantic descriptors, ECCVW, 2022.
[4] Kakogeorgiou et al., SPOT: Self-training with patch-order permutation for object-centric learning with autoregressive transformers, CVPR, 2024.
[5] Singh et al., Illiterate DALL-E learns to compose, ICLR, 2021.
[6] Tian et al., Pay attention to the foreground in object-centric learning, CVPR, 2025.

MUFASA


$\mathcal{S}_m$: Slots
$\hat{\mathcal{S}}_m$: Re-ordered slots
$\mathcal{A}_m^{\text{Slot}}$: Attention mask
① DINO
L_1
L_9
L_{10}
L_{11}
L_{12}
②
SA_1
SA_2
SA_3
SA_4
$\mathcal{S}_1$ $\mathcal{S}_2$ $\mathcal{S}_3$ $\mathcal{S}_4$
③
HM
HM
HM
$\hat{\mathcal{S}}_2$ $\hat{\mathcal{S}}_3$ $\hat{\mathcal{S}}_4$
Slot-wise Hungarian Matching
$\mathcal{A}_{\text{fused}}^{\text{Slot}}$
Fusion Module
$\mathcal{S}_{\text{fused}}$
Decoder
⑤
$\mathcal{A}^{\text{Dec}}$
⑥
$\mathcal{L}_{\text{Rec}}$
④

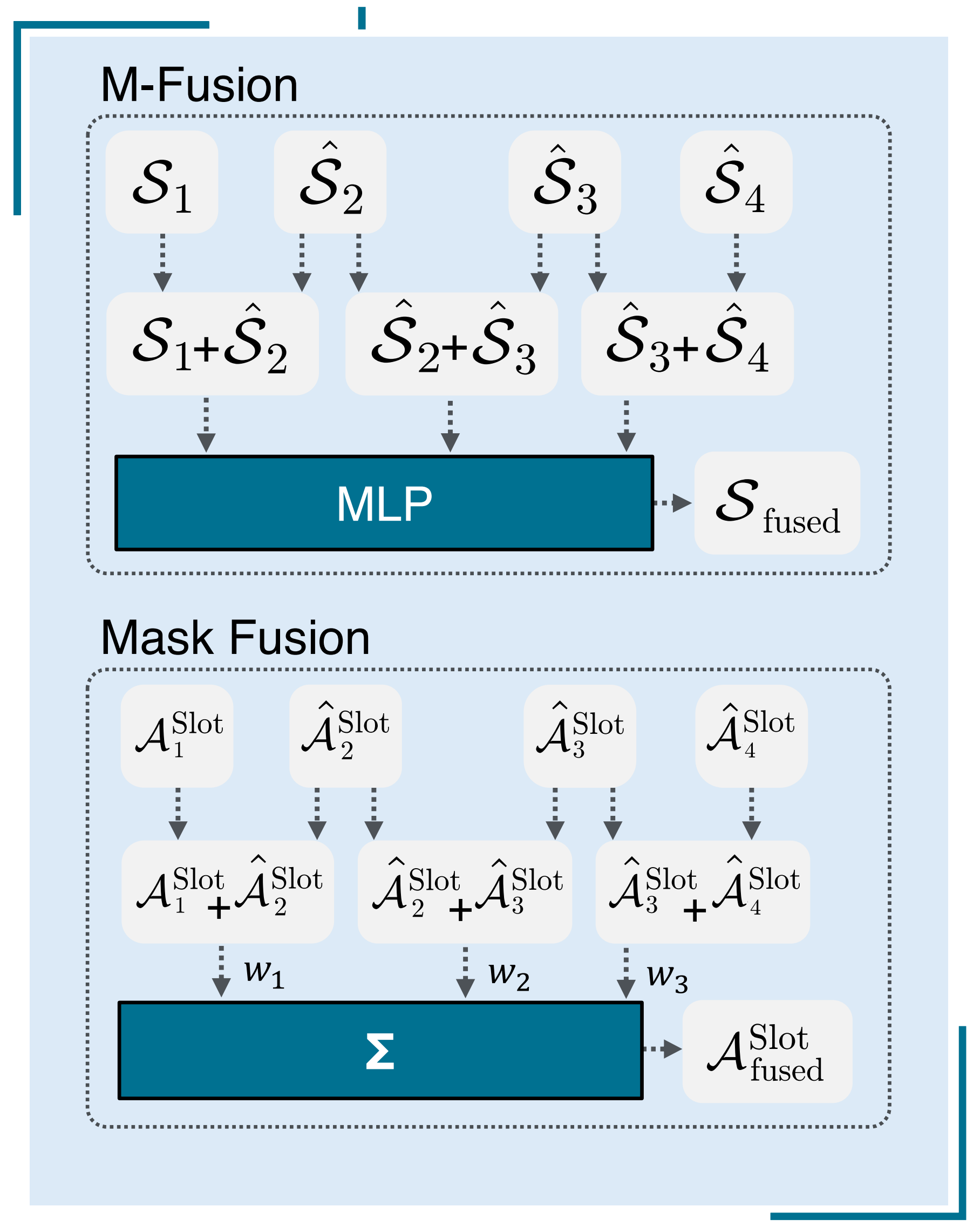
① The input image is processed by a **pre-trained DINO encoder**, from which feature representations are extracted at a **selected subset of encoder layers**.

② For the representation at each selected layer m, **slot attention** (SA) is performed through an independent SA module, yielding the set of slots $\mathcal{S}_m$ and the corresponding slot attention masks $\mathcal{A}_m^{\text{Slot}}$.

③ **Hungarian Matching (HM):** The set of slots and attention masks are re-ordered for best correspondence across layers, so that slots with the same index across layers learn to bind to the same object.

④ **Fusion Module:** Integrates multiple sets of slots and their attention masks into a unified representation:

- **Slot fusion:** The re-ordered sets of slots are summed in adjacent pairs, then concatenated and projected into a fused representation $\mathcal{S}_{\text{fused}}$ via an MLP.
- **Mask fusion:** The slot attention masks are summed and concatenated analogously, then fused using a weighted linear combination into $\mathcal{A}_{\text{fused}}^{\text{Slot}}$.


M-Fusion
$\mathcal{S}_1$ $\hat{\mathcal{S}}_2$ $\hat{\mathcal{S}}_3$ $\hat{\mathcal{S}}_4$
$\mathcal{S}_1+\hat{\mathcal{S}}_2$
$\hat{\mathcal{S}}_2+\hat{\mathcal{S}}_3$
$\hat{\mathcal{S}}_3+\hat{\mathcal{S}}_4$
MLP
$\mathcal{S}_{\text{fused}}$
Mask Fusion
$\mathcal{A}_1^{\text{Slot}}$ $\hat{\mathcal{A}}_2^{\text{Slot}}$ $\hat{\mathcal{A}}_3^{\text{Slot}}$ $\hat{\mathcal{A}}_4^{\text{Slot}}$
$\mathcal{A}_1^{\text{Slot}}+\hat{\mathcal{A}}_2^{\text{Slot}}$
$\hat{\mathcal{A}}_2^{\text{Slot}}+\hat{\mathcal{A}}_3^{\text{Slot}}$
$\hat{\mathcal{A}}_3^{\text{Slot}}+\hat{\mathcal{A}}_4^{\text{Slot}}$
w_1 w_2 w_3
Σ
$\mathcal{A}_{\text{fused}}^{\text{Slot}}$

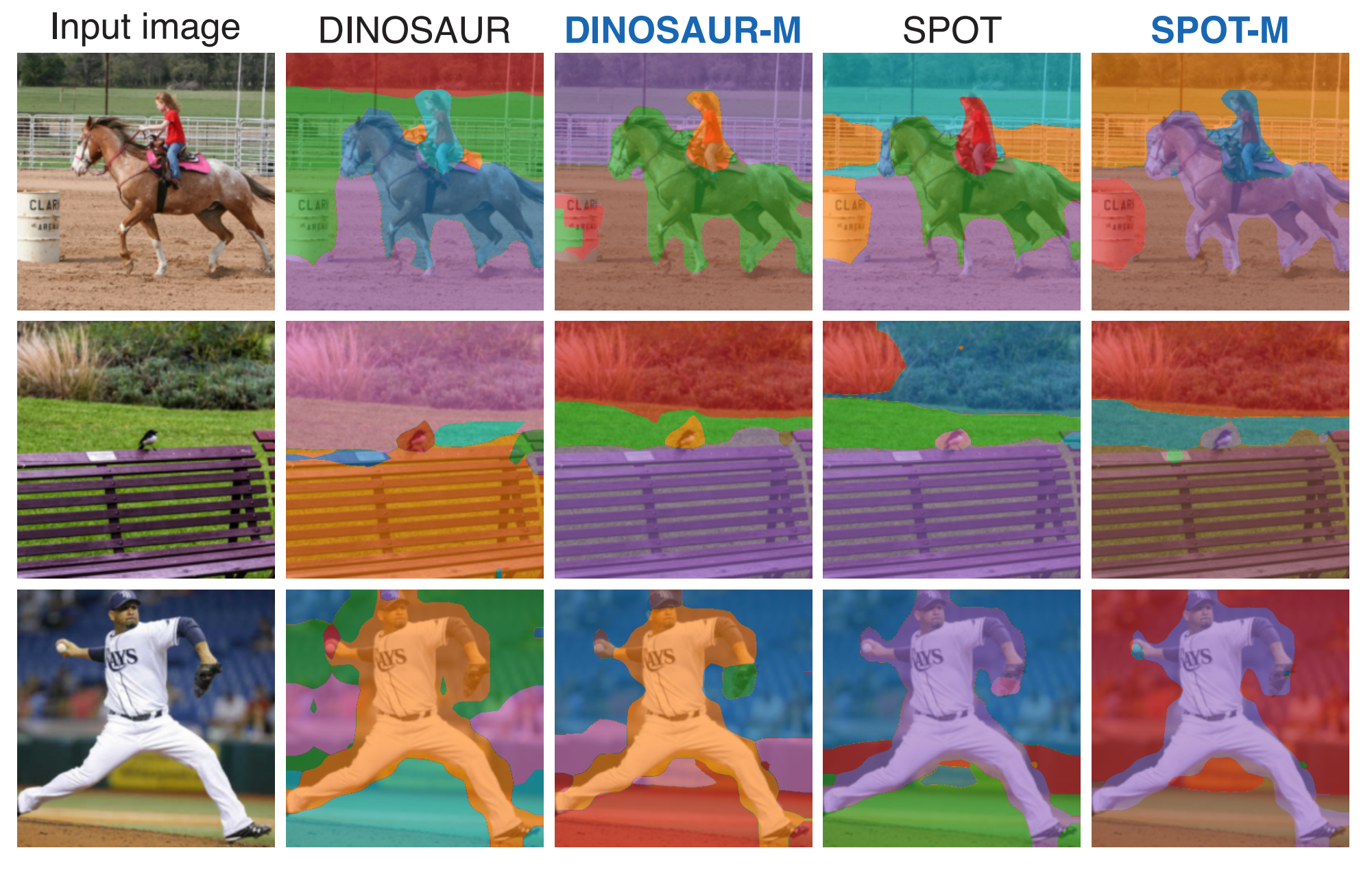
⑤ A **ViT decoder** reconstructs the last encoder layer's features from fused slots, yielding the decoder attention mask $\mathcal{A}^{\text{Dec}}$.

⑥ The reconstruction loss $\mathcal{L}_{\text{Rec}}$ guides the unsupervised training.

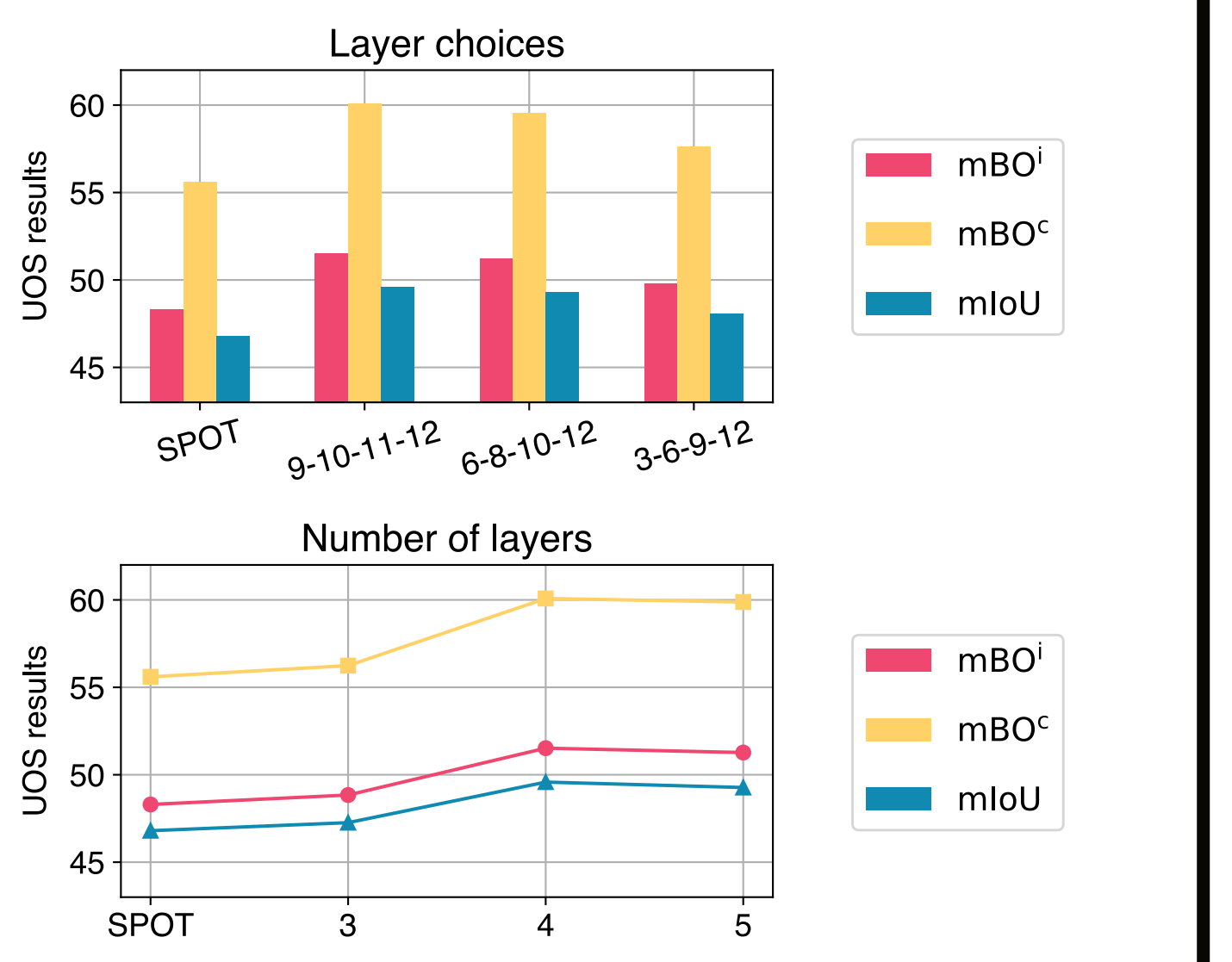
Results

Unsupervised object segmentation

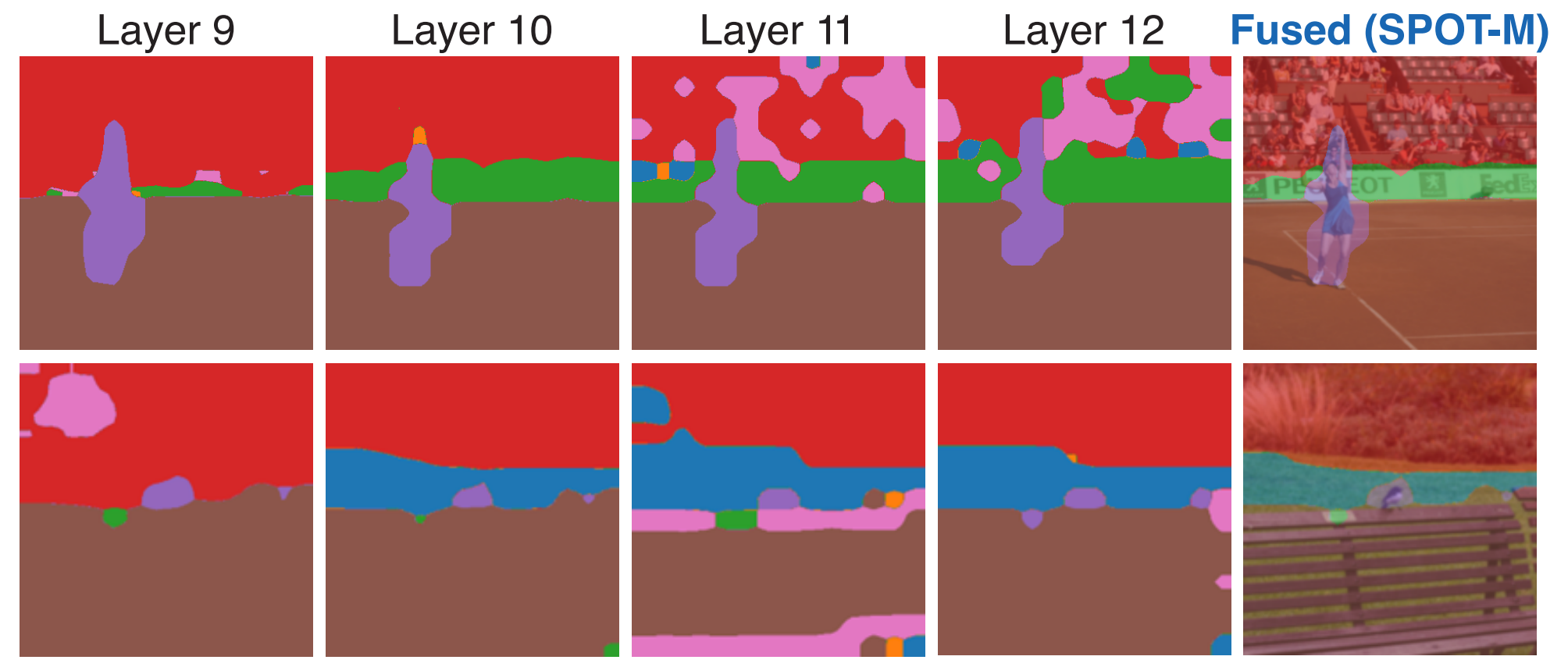
Model	Pascal VOC		COCO		MOVi-C	
	mBO^c	mBO^i	mBO^c	mBO^i	mBO^i	mIoU
Slot Attention [2]	24.9	24.6	19.2	17.2	26.2	–
SLATE [5]	41.5	35.9	33.6	29.1	39.4	37.8
SPOT-FS-RC [6]	56.5	49.3	45.3	**35.7**	49.0	47.8
DINOSAUR [1]	51.2 ↓ +6.4%	44.0 ↓ +5.2%	39.7 ↓ +3.3%	31.6 ↓ +1.1%	42.4 ↓ +6.8%	–
DINOSAUR-M *(ours)*	57.6	49.2	43.0	32.7	**49.2**	**48.3**
SPOT [4]	55.3 ↓ +4.5%	48.1 ↓ +3.2%	44.3 ↓ +1.2%	34.7 ↓ +0.1%	47.0 ↓ +2.2%	46.4 ↓ +1.8%
SPOT-M *(ours)*	**59.8**	**51.3**	**45.5**	34.8	**49.2**	48.2


Input image
DINOSAUR
DINOSAUR-M
SPOT
SPOT-M

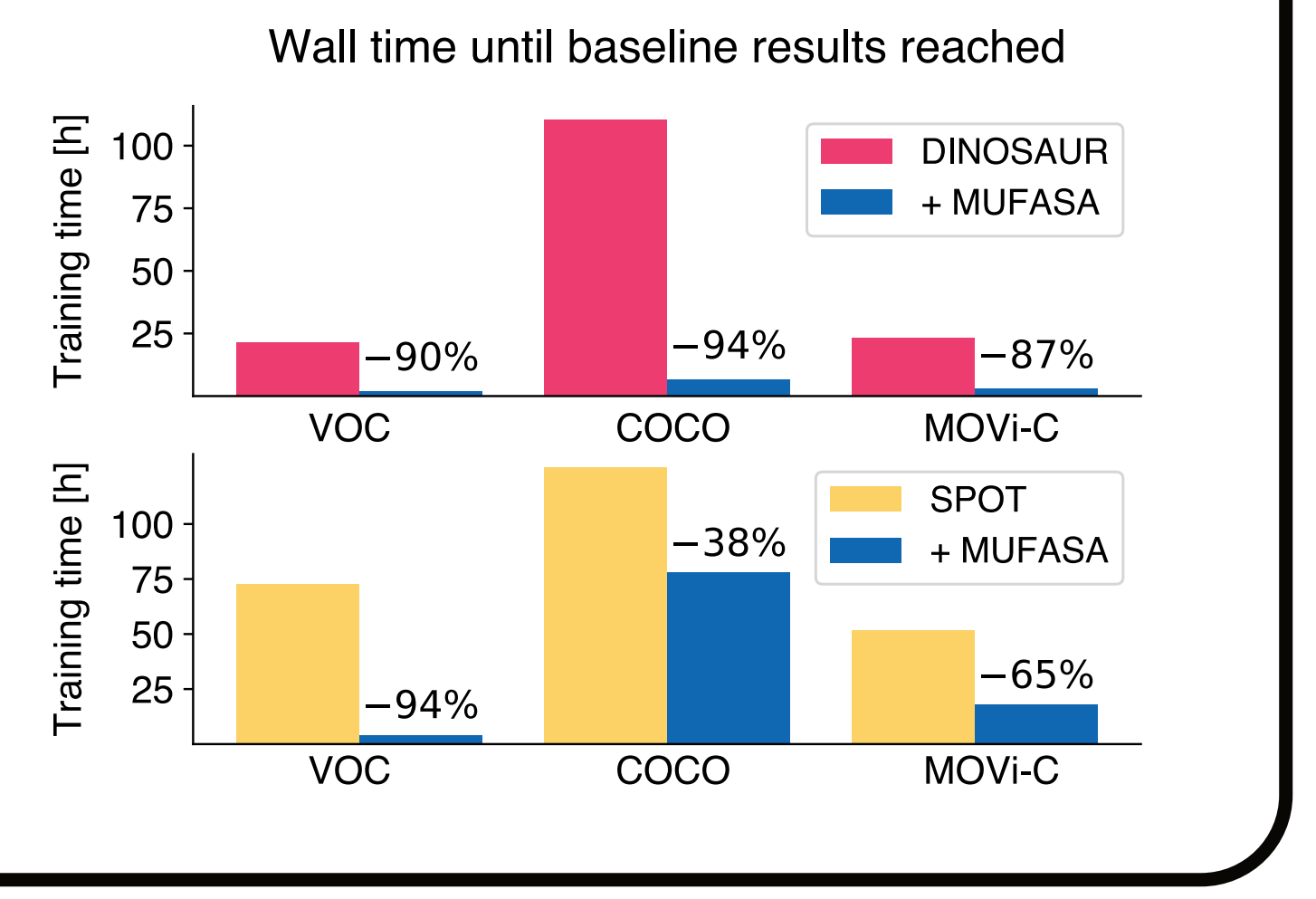
Layer selection



Layer-wise segmentations


Layer 9
Layer 10
Layer 11
Layer 12
Fused (SPOT-M)

Training efficiency



Takeaways

- Semantic information is not confined to the final layer of a pre-trained DINO encoder. MUFASA leverages this multi-layer information for unsupervised object segmentation.
- MUFASA can easily be applied to existing slot-attention methods utilizing a DINO encoder, improving their segmentation results.
- Integrated into SPOT, MUFASA sets a new state of the art for unsupervised object segmentation across multiple datasets.
- MUFASA substantially improves training convergence with only minor inference overhead.

Acknowledgements This project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 program (grant agreement No. 866008). The project was also supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy (EXC-3057/1 "Reasonable Artificial Intelligence", Project No. 533677015). Simone Schaub-Meyer has been funded by the DFG - 529680848. Leonie Schüßler and Sebastian Bock are supported by the Konrad Zuse School of Excellence in Learning and Intelligent Systems (ELIZA) through the DAAD programme Konrad Zuse Schools of Excellence in Artificial Intelligence, sponsored by the German Federal Ministry of Education and Research.

erc European Research Council

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